UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

DECEMBER EXAMINATION

UCCM1363 DISCRETE MATHEMATICS

SATURDAY, 30 DECEMBER 2023 TIME: 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)

Instructions to Candidates :

Answer **ALL** questions in **section A.**

Answer **ANY ONE** question in **section B.**

If more than **ONE** question from **section B** is answered, then only the **first question attempted in section B will be marked.**

All questions carry equal marks. Marks allocated for each part of the questions are indicated in brackets.

SECTION A

Answer ALL questions.

Q1. (a) Give the negation, converse, inverse and contrapositive of the statement.

If Jason buys the book, then Jason goes to the shop.
(8 marks)

(b) Sara is a math major, or Sara is an economics major.
If Sara is a math major, then Sara is required to take Calculus.
Therefore, Sara is an economics major or Sara is not required to take Calculus.

Based on the given argument form, write the symbolic form, and then determine its validity by using truth table. (10 marks)

(c) Given the statement is $\forall x,y$ in $\mathbb{R}$ such that $\sqrt{x+y} = \sqrt{x} + \sqrt{y}$.

(i) Determine the truth value of the statement and give explanation for the answer. (3 marks)

(ii) Give the negation of the statement in formal and informal statements.
(4 marks)
[Total: 25 marks]

Q2. (a) Guess an explicit formula for the following sequence with initial terms given.
$\frac{1}{4}, -\frac{2}{9}, \frac{3}{16}, -\frac{4}{25}, \frac{5}{36}, -\frac{6}{49}.$ (3 marks)

(b) Use mathematical induction to prove that $n^3 - 7n + 3$ is divisible by 3, for each integer $n \geq 0$. (10 marks)

(c) The Fibonacci sequence satisfies the recurrence relation $F_k = F_{k-1} + F_{k-2}$, for all integers $k \geq 2$.

(i) Write an equation expressing F_{k+3} in terms of F_{k+2} and F_{k+1} and find Fibonacci number F_3, F_4, F_5, and F_6 if $F_1 = 1$ and F_2=2. (6 marks)

(ii) Prove that $F_k^2 - F_{k-1}^2 = F_k F_{k+1} - F_{k-1} F_{k+1}$ for all integers $k \geq 1$. (6 marks)
[Total: 25 marks]

Q3. (a) Let $A = \{1, 2, 3, 4, 5\}$ and R be the relation on A defined by

$$\begin{pmatrix} 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{pmatrix}$$

(i) Find the quotient set of A. (4 marks)

(ii) With explanation, determine whether the relation is reflexive, irreflexive, symmetric, asymmetric, antisymmetric, or transitive. (12 marks)

(b) Consider set $B = \{4, 8, 12, 16, 36, 144, 288, 432\}$ and its subset $B_1 = \{8, 12\}$.

(i) Draw the Hasse diagram where the poset is $(B,|)$. (5 marks)

(ii) Determine the maximal and minimal elements of B, if they exist. (2 marks)

(iii) Determine the greatest and least elements of B, if they exist. (2 marks)

[Total: 25 marks]

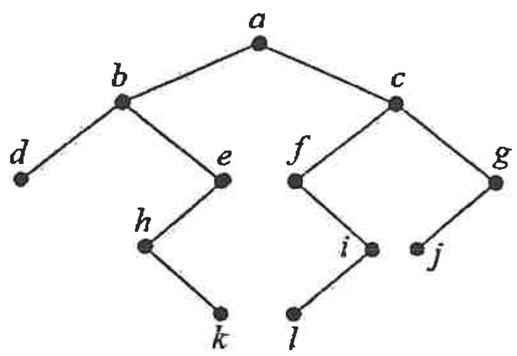
SECTION B: Answer only **ONE** question.

Q4. (a) Draw a labeled tree to represent the following fully parenthesized algebraic expression:

$(3 \times (1 - x)) \div ((4 + (7 - (y + 2))) \times (7 + (x \div y)))$. (4 marks)

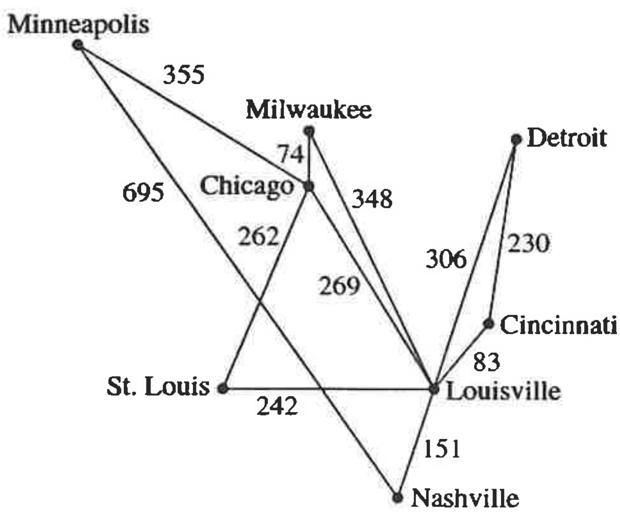
(b) Write the specific orders followed in the process of visiting each vertex of a tree for preorder search, inorder search and postorder search. (6 marks)

(c) Perform preorder search, inorder search, and postorder search of the tree below:



(15 marks)
[Total: 25 marks]

Q5. (a) Find the minimal spanning tree for the diagram below by using:



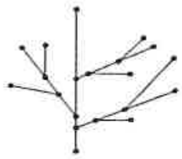
(i) Kruskal's algorithm. (10 marks)

(ii) Prim's algorithm using the Minneapolis vertex as a starting point. (9 marks)

Q5. (Continued)

(b) Determine the following graphs are trees or not and give your reason.

(i)

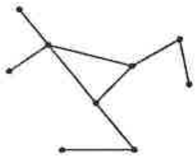


(2 marks)

(ii)

(2 marks)

(iii)



(2 marks)
[Total: 25 marks]

UCCM1363 DISCRETE MATHEMATICS

Formula Sheet

Let p and q be statement variables:
(1) The **negation** of $p \Rightarrow q$ is $p \wedge \sim q$.
(2) The **contrapositive** of $p \Rightarrow q$ is $\sim q \Rightarrow \sim p$.
(3) The **converse** of $p \Rightarrow q$ is $q \Rightarrow p$.
(4) The **inverse** of $p \Rightarrow q$ is $\sim p \Rightarrow \sim q$.

Theorem 1.1
Let p and q be statements variables.
(1) $p \Rightarrow q \equiv \sim p \vee q$
(2) $p \Rightarrow q \equiv \sim q \Rightarrow \sim p$
(3) $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$

Theorem 1.2
(1) Commutative laws:
$p \wedge q \equiv q \wedge p$
$p \vee q \equiv q \vee p$

(2) Associative laws:
$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
$(p \vee q) \vee r \equiv p \vee (q \vee r)$

(3) Distributive laws:
$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

(4) Identity laws:
$p \wedge t \equiv p$
$p \vee c \equiv p$

(5) Negation laws:
$p \vee \sim p \equiv t$
$p \wedge \sim p \equiv c$

(6) Double negative laws:
$\sim(\sim p) \equiv p$

(7) Idempotent laws:
$p \wedge p \equiv p$
$p \vee p \equiv p$

(8) Universal bound laws:
$p \vee t \equiv t$
$p \wedge c \equiv c$

(9) De Morgan's laws:
$\sim(p \wedge q) \equiv \sim p \vee \sim q$
$\sim(p \vee q) \equiv \sim p \wedge \sim q$

(10) Absorption laws:
$p \vee (p \wedge q) \equiv p$
$p \wedge (p \vee q) \equiv p$

(11) Negations of t and c:
$\sim t \equiv c$
$\sim c \equiv t$

	Rules of Inference	Name
1.	$p \Rightarrow q$ p $\therefore q$	Modus Ponens (Law of Detachment)
2.	$p \Rightarrow q$ $\sim q$ $\therefore \sim p$	Modus Tollens
3.	p or q $\therefore p \vee q$ $\therefore p \vee q$	Disjunctive Addition
4.	$p \wedge q$ or $p \wedge q$ $\therefore p$ $\therefore q$	Conjunctive Simplification
5.	$p \vee q$ or $p \vee q$ $\sim p$ $\sim q$ $\therefore q$ $\therefore p$	Disjunctive Syllogism
6.	p q $\therefore p \wedge q$	Conjunction
7.	$p \Rightarrow q$ $q \Rightarrow r$ $\therefore p \Rightarrow r$	Hypothetical Syllogism
8.	$p \vee q$ $\sim p \vee r$ $\therefore q \vee r$	Resolution